

Q-MODE: Amino Acid Lattice Mapping for Binding-Pocket Representation and Quantum-Assisted Docking Site Search

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Abstract

Q-MODE is a pipeline that converts a protein, provided as a PDB structure, in a discrete sequence of pharmacophore interactions action sites maintaining spatial coherence and subsequently encodes that sequence into qubit-ready quantum states. The pipeline combines chemistry computer science tools such as RDKit, Crippen and Intensities derived from Abraham solvation with a coding step that exploits quantum theory, following the two-step scheme (first/second encoding) and the modified formulation of Grover’s search introduced in “Quantum algorithm for identifying protein-ligand docking sites in interaction space”. This report documents the full operation of the pipeline and its evaluation on 29 crystallographic protein–ligand complexes. The end-to-end result is negative: the closest window the representation makes available sits at a median 3.76 Å from the true ligand centroid and a window drawn at random at 7.88 Å, while the search returns 7.75 Å. We show that the criterion was attainable, that ranking, window size, matching regime and the 3D-to-1D flattening are not responsible, and that the cause is a descriptor which records the composition of a surface patch but not its enclosure.

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1 Introduction

Identifying the binding site of a ligand on a protein surface is a central problem in structure-based drug design. When a candidate drug molecule approaches a protein, it binds preferentially to a specific region — the binding pocket — where its chemical properties are complementary to those of the protein surface. Predicting which region of the pocket is chemically compatible with a given ligand is computationally expensive with classical methods, motivating the search for quantum approaches that could offer a computational advantage.

Q-MODE addresses this problem by converting the protein into a discrete, ordered sequence of pharmacophoric interaction sites — a flat 1D chain analogous to a sequence of residues in a protein. Each site in the chain carries two scalar values that quantify its hydrophobic character and its hydrogen-bonding capacity. Short windows of this chain, of the same length as the ligand’s own pharmacophoric profile, are then encoded into low-qubit-count quantum states using the two-step scheme introduced by Liliopoulos et al. This representation is directly compatible with amplitude encoding and basis-state encoding schemes used by quantum search algorithms — in particular, a modified Grover search — which scan the protein chain for windows whose encoded profile matches that of the ligand.

1.1 Scientific Background

The quantum-encoding and quantum-search stages of Q-MODE follow the scheme proposed in “*Quantum algorithm for protein-ligand docking sites identification in the interaction space*” (Liliopoulos et al.):

- A **first encoding** that binarizes, per site, a hydrophobicity channel (h) and a hydrogen-bonding/charge channel (hb) into a 2-qubit basis state, so that a window of k sites maps onto a $2k$ -qubit computational basis state. This is the representation used to build the Grover oracle.
- A **second encoding** that maps the same two channels onto two pairs of probability amplitudes (a, b) and (c, d), suitable for amplitude-based (e.g. SWAP-test) similarity/distance estimation between two sites or windows.
- A **modified Grover search**, performed for “protein shift”, which identifies which windows’ encode bit- for the first time the string corresponds to the bit string of the ligand with probability above the uniform threshold ($1/N$).

This project implements the first encoding, the second encoding, and the modified Grover search.

2 Pipeline Overview

Given a protein PDB file, Q-MODE executes the following stages in sequence:

1. **Residue extraction** — the PDB file is parsed line by line to extract real 3D atomic coordinates from the crystal structure. For each amino acid, an RDKit molecule is built with the correct chemical topology. Non-standard residues, water molecules and ligands are discarded (Section 3.1).
2. **Pharmacophoric feature extraction** — RDKit’s Feature Factory is applied to each residue using the BaseFeatures.fdef file, which contains SMARTS rules for recognising six pharmacophore interaction types: HBondDonor, HBondAcceptor, Hydrophobe, Aromatic, PosIonizable and NegIonizable. Each extracted feature is assigned a 3D position corresponding to the centroid of its constituent atoms (Section 3.2).

3. **Intensity assignment** — hydrophobic site intensities are computed from per-atom partial LogP contributions using the Crippen method. For hydrogen-bond donor, acceptor and ionizable sites, intensities are derived from Abraham solute descriptors — empirical parameters that quantify the interaction capacity of each functional group (Section 3.3).
4. **Surface filtering** — per-atom SASA (Solvent Accessible Surface Area) is computed for the protein structure using the Shrake & Rupley algorithm with a spherical probe of radius 1.4 Å representing a water molecule. The calculation is performed on the ligand-free protein to prevent the co-crystallised ligand from occluding the binding pocket. Pharmacophoric features whose representative atoms fall below the SASA threshold are discarded as physically inaccessible to a docking ligand (Section 3.4).
5. **Site deduplication** — features of the same pharmacophore type located within 1.5 Å of each other are merged into a single representative site whose coordinates are the centroid of the group. Features of the same type that are geometrically distant are retained as independent sites with their original coordinates (Section 3.5).
6. **Topological ordering** — the pharmacophoric sites of each residue are reordered according to the covalent connectivity of the molecule via a breadth-first traversal of the bond graph, starting from the backbone nitrogen. This ensures that the site order reflects the chemical structure of the residue — backbone first, then the side chain from its attachment point outward (Section 3.6).
7. **Chain assembly** — the sites of all residues are concatenated into a single flat sequence following protein chain order — sorted by chain ID and ascending residue sequence number. The result is the flat chain: an ordered 1D sequence of pharmacophoric sites carrying their 3D coordinates, types and intensities, ready for quantum encoding (Section 3.7).
8. **Quantum encoding** — first (binary) and second (amplitude) encodings of ligand-sized sliding-window segments (Section 3.8).
9. **Ligand extraction** — parsing of a bound ligand’s HETATM group and construction of its own pharmacophore chain (Section 3.9).
10. **Grover search** — quantum-circuit-based identification of candidate docking windows matching the ligand’s encoded profile (Section 3.10).
11. **Distance-to-ligand validation** — see Section 3.11

Figure 1 sketches the data flow between stages.

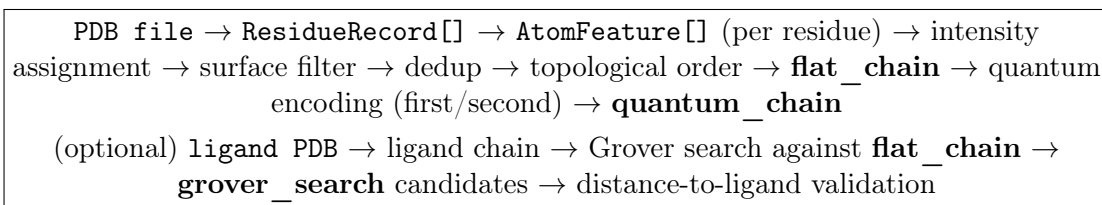


Figure 1: High-level data flow of the Q-MODE pipeline.

3 Methods

3.1 Residue Extraction and Order Assignment

Each residue is extracted directly from the PDB file by reading the **ATOM** and **HETATM** records. When multiple alternate conformations are present, only the primary one is kept. Water molecules

are discarded by default. Residues are grouped by their chain identifier, sequence number and residue name, then sorted by chain and sequence number to preserve the natural order of the protein chain.

For each residue, an RDKit molecule is constructed from the real heavy-atom coordinates read directly from the PDB. Bond orders and aromaticity are then assigned by comparing the residue against a canonical SMILES template for each of the 20 standard amino acids. After bond assignment, missing hydrogens, which are typically not resolved by X-ray crystallography, are added with 3D coordinates consistent with the heavy-atom geometry (`AddHs(addCoords=True)`).

3.2 Pharmacophoric Feature Extraction

Each residue’s RDKit molecule is analysed by RDKit’s chemical feature factory, which uses the `BaseFeatures.fdef` rule file to identify pharmacophoric groups based on SMARTS pattern matching. The factory recognises six interaction families, which are mapped onto the internal type vocabulary used throughout Q-MODE:

- `HBondDonor`
- `HBondAcceptor`
- `Hydrophobe`
- `Aromatic`
- `PosIonizable`
- `NegIonizable`

Each extracted feature carries the 3D position of its representative point, the centroid of its constituent atoms, along with the list of atom indices involved. This index list is used by subsequent pipeline stages for intensity assignment, surface filtering and topological ordering. For multi-atom groups such as aromatic rings or guanidinium groups, the centroid is computed as the unweighted mean of the constituent heavy-atom positions.

3.3 Intensity Assignment

Every pharmacophoric site carries a scalar *intensity* intended to reflect the physical strength of its interaction potential. Two separate methods are used depending on the site type.

For hydrophobic sites, the intensity is derived from Crippen’s method for per-atom LogP decomposition. The coefficient LogP measures how much a molecule prefers a non-polar environment over water; Crippen’s approach decomposes this global property into additive per-atom contributions by classifying each heavy atom. A positive contribution indicates a hydrophobic atom; a negative one indicates a polar or charged atom. The intensity of a `Hydrophobe` site is set to the sum of positive Crippen contributions over its constituent atoms, falling back to a neutral value of 1.0 when none are positive. This produces a continuous measure of local hydrophobic density rather than a binary label.

For hydrogen-bond donor, acceptor and ionizable sites, intensities are drawn from Abraham’s Linear Solvation Energy Relationship (LSER) framework. Abraham’s α_2^H descriptor quantifies the effective hydrogen-bond donor strength of a site, that is, how strongly it can donate a hydrogen to an acceptor, while β_2^H quantifies the acceptor strength. Both are dimensionless, non-negative, and derived from experimental measurements of partition coefficients across solvent systems with known hydrogen-bonding properties. In Q-MODE, α_2^H values are assigned to `HBondDonor` and `PosIonizable` sites, and β_2^H values to `HBondAcceptor` and `NegIonizable` sites. Values are looked up in a table keyed by residue name. When no exact entry is found, as is

always the case for ligand atoms, which do not match any standard amino-acid key, a heuristic classifier inspects the atom’s local chemical environment (element, valence, neighbouring atoms) and maps it to the closest available proxy among four functional-group classes: carboxyl, amine, phenol and thiol. Atoms outside these four classes retain the neutral default intensity.

3.4 Surface Filtering

Buried sites are not physically reachable by an incoming ligand and would otherwise add useless information to the protein representation. To remove them we apply a surface accessibility filter based on the Solvent-Accessible Surface Area (SASA), a measure of how much of each atom’s surface can be touched by a water-sized spherical probe rolling over the protein surface. An atom that is fully surrounded by other atoms has $SASA = 0$ and is considered buried; an atom on the protein surface has $SASA > 0$ and is considered exposed. This concept was introduced by Lee & Richards and the numerical algorithm used here is that of Shrake & Rupley [4], implemented in BioPython with a probe radius of 1.4 Å, the approximate radius of a water molecule.

The calculation is performed on the *ligand-free* protein structure: any co-crystallised ligands, water molecules and ions are stripped before computing SASA.

A pharmacophoric site is retained if its representative atom, the PDB atom geometrically closest to the feature centroid, has $SASA > \tau$, where the default threshold is $\tau = 1.0 \text{ Å}^2$. The literature typically considers values in the range 1.0–5.0 Å² as the threshold for significant exposure. The filter is active by default and can be disabled with the `-no-surface-filter` flag, for example when buried sites are of interest for structural comparison rather than ligandability analysis.

Because RDKit atom indices can become misaligned with PDB atom names during template-based bond assignment, the association between a pharmacophoric feature and its SASA value is established by nearest 3D coordinate rather than by atom index — Coordinate-based matching is used because RDKit atom indices can become misaligned with PDB atom names during template-based bond assignment (Section 3.1). This approach is more robust to the index misalignment that can arise from the template matching procedure described in Section 3.1.

A fundamental limitation of the SASA-based filter is that it measures solvent accessibility globally, without distinguishing between atoms exposed towards the interior of the binding pocket, where a docking ligand would interact, and atoms exposed towards the bulk solvent on the outer surface of the protein. As a consequence, pharmacophoric sites located on solvent-exposed loops or surface helices far from the binding site may pass the filter despite being irrelevant for ligand binding, while sites on the inner walls of a deep or partially occluded pocket may be incorrectly discarded.

3.5 Site Deduplication

Within a residue, several nearby atoms of the same pharmacophore type would otherwise appear as separate, highly-correlated sites. `site_dedup_centroid.py` greedily groups same-type features whose pairwise Euclidean distance is below 1.5 Å, replacing each group with a single site at the group’s coordinate centroid and an aggregate intensity equal to the *sum* of the group’s individual intensities.

3.6 Topological Ordering

Once the pharmacophoric sites of a residue have been deduplicated, they are reordered to follow the chemical connectivity of the residue. This step is necessary because feature extraction and deduplication produce sites in an order that depends on the internal implementation of RDKit rather than on the chemical structure of the residue. A reproducible, chemically motivated ordering is required for the flat chain to be stable across pipeline runs on the same input.

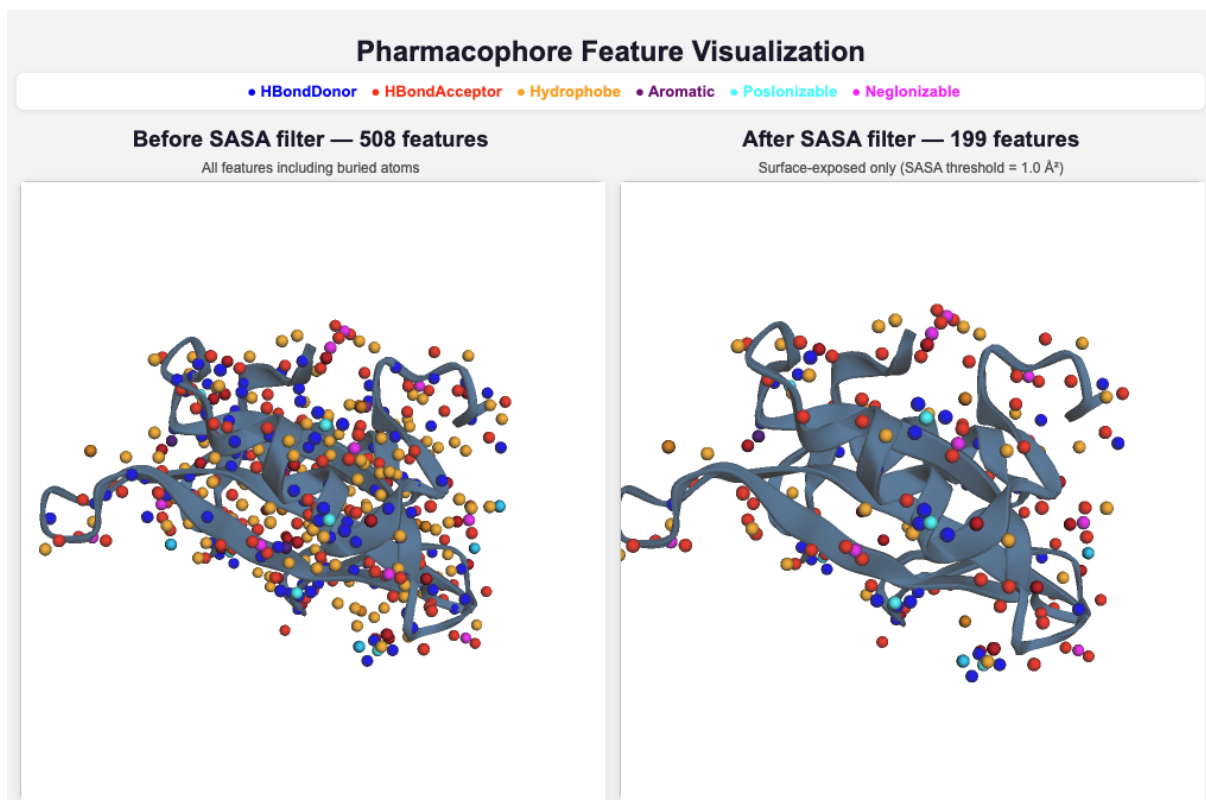


Figure 2: Enter Caption

Figure 3: Pharmacophoric sites before and after SASA surface filtering. Left: all 508 extracted sites including buried atoms. Right: 199 surface-exposed sites retained after applying the SASA filter with threshold $\tau = 1.0 \text{ \AA}^2$ (protein 1a08). The counts are those of Table ??.

The ordering is computed by a breadth-first search (BFS) over the residue’s covalent bond graph, in which nodes are the heavy atoms and edges are the covalent bonds. BFS visits all nodes at topological distance 1 from the starting node before moving to distance 2, and so on, producing a visit order that reflects proximity in the bond graph rather than in 3D space. The traversal starts from the heavy atom with the lowest RDKit index and neighbours are visited in sorted index order at each step to guarantee determinism. Each pharmacophoric site is associated with the nearest heavy atom by Euclidean distance, and the BFS visit order of that representative atom determines the site’s position in the residue-local sequence.

Once all residues have been processed individually, their locally ordered site lists are concatenated in ascending chain and sequence-number order to produce the final flat chain. The ordering is therefore two-level: BFS within each residue enforces chemical coherence at the local scale, while chain order preserves the primary structure of the protein at the global scale.

Validation

The ordering step was validated on four structures using two criteria. First, determinism: running the BFS twice on the same residue always produces the same sequence, verified across all residues of all four structures with zero exceptions. This confirms that sorted neighbour traversal and a fixed starting atom are sufficient to make the ordering fully reproducible.

Second, the fraction of residues whose first site is associated with a backbone atom (N, CA, C or O) was measured across all four structures (Table 2). Between 63% and 77% of residues satisfy this criterion.

As a concrete illustration, residue A159_ARG in 1a08 produces six sites after surface filtering

Table 1: Effect of the SASA surface filter and site deduplication across four protein structures. Reduction percentages are computed relative to the previous stage.

Protein	Feature extraction	After SASA	After dedup	SASA red.	Dedup red.	Total red.
1a08	508	199	199	−60.8%	−0.0%	−60.8%
1a0t	6015	1926	1911	−68.0%	−0.8%	−68.2%
1a1c	1013	378	377	−62.7%	−0.3%	−62.8%
1a46	1448	774	756	−46.5%	−2.3%	−47.8%

Table 2: Topological ordering validation: backbone-first rate across four protein structures. All structures are fully deterministic.

Protein	Residues	Backbone first	Rate
1a08	52	36	69.2%
1a0t	487	306	62.8%
1a1c	103	69	67.0%
1a46	181	140	77.3%

and deduplication, occupying positions 37–42 in the flat chain (Table ??). The ordering proceeds from the backbone HBondAcceptor through the hydrophobic $C\beta$ and then along the guanidinium group in bond-graph order, confirming that the BFS correctly captures the chemical topology of the residue.

3.7 Chain Assembly

Residues are concatenated in protein-chain order — i.e. by `(chain_id, res_seq)` as already established at the parsing stage — and each residue’s internally-ordered sites are appended in sequence. The result is one flat list, `flat_chain`, where every entry carries: the originating residue label, residue name/sequence/chain id, 3D coordinates, pharmacophore type, and (channel-appropriate) intensity. This flat chain is the shared representation consumed by both the visualization utilities and the quantum-encoding stage.

3.8 Quantum Encoding

The flat chain produced by the pre-processing pipeline is the input to the quantum encoding stage, which translates the pharmacophoric site sequence into quantum states suitable for the Grover-based docking search described in Lillipoulos et al. Each site is first projected onto two scalar interaction channels, and then encoded twice — once as a discrete basis state for the quantum search, and once as a continuous amplitude state for similarity evaluation.

Channel projection. Each site in `flat_chain` is reduced to two scalar channels (function `get_h_hb_intensities` in `qubit_chain.py`): a hydrophobic channel h and a hydrogen-bonding channel hb :

$$(h, hb) = \begin{cases} (\text{intensity}, 0) & \text{type} \in \{\text{Hydrophobe}, \text{Aromatic}\} \\ (0, \text{intensity}) & \text{type} \in \{\text{HBondDonor}, \text{HBondAcceptor}, \text{PosIonizable}, \text{NegIonizable}\} \end{cases}$$

This separation reflects the two interaction types considered by the quantum algorithm: hydrophobic contacts and hydrogen bonds. Aromatic is routed to h because a ring face presents an apolar contact surface, while PosIonizable/NegIonizable are routed to hb , rather than dropped to

(0,0), because the dominant directional contact of a charged group is a charge-assisted hydrogen bond.

First encoding — basis-state binarization. Each channel value is binarized against a per-channel threshold, producing a 2-bit basis state per site:

$$q_h = \begin{cases} 0 & h < h_{\text{thr}} \\ 1 & h \geq h_{\text{thr}} \end{cases} \quad q_{hb} = \begin{cases} 0 & hb < hb_{\text{thr}} \\ 1 & hb \geq hb_{\text{thr}} \end{cases}$$

The reference paper prescribes thresholding at the channel’s range midpoint $(v_{\min} + v_{\max})/2$. However, pharmacophoric intensity distributions are typically right-skewed, the majority of sites cluster near the lower bound, with a small number of high-intensity outliers. Under these conditions, the midpoint threshold classifies nearly all sites as 0, producing an uninformative near-zero bitstring. Q-MODE therefore thresholds at the *median* of each channel’s non-zero values (`compute_h_hb_thresholds`), which splits the active population roughly evenly and preserves the discriminative power of the binarization for the downstream Grover search.

A sliding window of k consecutive sites, where k equals the number of interaction sites of the ligand, is mapped onto a $2k$ -bit basis state:

$$|x\rangle = \left| q_h^{(1)} q_{hb}^{(1)} \cdots q_h^{(k)} q_{hb}^{(k)} \right\rangle$$

Second encoding — amplitude encoding. Each channel value v is also encoded as a pair of normalised probability amplitudes by projecting v within its channel range $[v_{\min}, v_{\max}]$:

$$p = v_{\max} - v, \quad q = v - v_{\min}, \quad (a, b) = \left(\frac{p}{\sqrt{p^2 + q^2}}, \frac{q}{\sqrt{p^2 + q^2}} \right)$$

Applied independently to h (yielding amplitudes a, b with $a^2 + b^2 = 1$) and to hb (yielding c, d with $c^2 + d^2 = 1$). The value v is clamped to $[v_{\min}, v_{\max}]$ before projection to ensure that sites with zero contribution on a channel.

This amplitude representation encodes the continuous intensity information of each site as a quantum state, enabling the SWAP-test-based Euclidean distance estimation between protein and ligand site sequences prescribed by the reference paper.

Sliding-window segmentation. `qubit_chain.build_qubit_chain` slides a window of size `ligand_size` (default 3 sites) over `flat_chain` with unit stride, producing one `QubitSegment` per window position, each carrying its first-encoding bitstring and its per-site second-encoding amplitudes. This is the “classical” quantum-encoding step (Step 8 in the pipeline enumeration), independent of the real ligand used later by Grover search.

3.9 Ligand Extraction

The quantum docking search requires a ligand-side representation of the same form as a protein window, a short sequence of pharmacophoric sites carrying hydrophobic and hydrogen-bonding intensities. When a full PDB structure containing a bound ligand is supplied via `-ligand-pdb`, the ligand is extracted and processed through the same local pipeline applied to each protein residue, yielding a ligand chain that can be directly compared to protein windows in the Grover search stage.

The ligand reader (`ligand_reader.py`) works on the HETATM lines of the input file, the lines a PDB uses for everything that is not part of the protein chain. It reads them one at a time and throws anything whose residue name is one of the 20 standard amino acids. What is left is grouped by the triple (`chain_id, res_seq, res_name`). Atoms read from a PDB file come

with positions but with no bonds, so the bonds have to be filled in before the molecule can be analysed. Two routes are tried, in order. The first looks up the ligand’s three-letter code in the RCSB Chemical Component Dictionary, an online catalogue holding a reference description of the molecule behind each code, and copies the bonds of that reference onto the atoms read from the file. The second is used only when the first fails, either because the code is not in the catalogue or because there is no network: the bonds are then worked out from the coordinates alone, joining atoms that sit close enough to be bonded and choosing single or double so that every atom ends up with a plausible number of bonds.

From here the ligand is treated as one more residue, with no special case: the same feature extraction, the same intensity assignment, the same merging of nearby features of the same type (capped at `-ligand-max-sites`, default 3), and the same ordering along the bonds. The output is a short sequence of sites carrying the same two numbers h and hb per site as a protein window, which is exactly what the encoding and search stages compare.

3.10 Grover Search

Once the protein flat chain and the ligand chain have been encoded, Q-MODE searches for protein windows whose pharmacophoric pattern matches the ligand. This is formulated as a database search problem: the protein windows are the database entries, the ligand bitstring is the target element, and Grover’s quantum search algorithm is used to amplify the probability of measuring the matching entry. The key quantum advantage is that Grover’s algorithm finds the target in $O(\sqrt{N})$ oracle evaluations rather than the $O(N)$ required by classical exhaustive search, where N is the number of unique protein windows.

For a ligand chain of k sites, Q-MODE runs a modified Grover search (`qmode/grover/search.py`) against every non-overlapping tiling of the protein’s `flat_chain` into windows of size k .

Tiling. For each shift offset $o \in \{0, \dots, k-1\}$, the pocket chain is partitioned into non-overlapping windows of size k starting at position o (`tile_offset`). Each window is reduced to its first-encoding bitstring. When several windows share the same bitstring, the one with the highest aggregate h/hb intensity — the most chemically interactive window — is retained as that bitstring’s representative.

Superposition and oracle. The unique bitstrings observed at offset o are loaded into an equal superposition over $n_{\text{qubits}} = 2k$ qubits:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^N |x_i\rangle,$$

where N is the number of unique windows at this offset. The ligand’s first-encoding bitstring x_{lig} defines a phase-flip oracle:

$$\hat{O} = I - 2|x_{\text{lig}}\rangle\langle x_{\text{lig}}|,$$

which negates the amplitude of the target state while leaving all others unchanged. The Grover diffusion operator:

$$\hat{G} = 2|s\rangle\langle s| - I,$$

is then applied to invert all amplitudes about the mean, amplifying the target state’s probability above the uniform baseline $1/N$.

Circuit execution. The superposition state $|s\rangle$ is prepared with Qiskit’s `StatePreparation`, followed by the oracle and diffusion operators compiled from their dense unitary matrices via `UnitaryGate`. The circuit is transpiled and executed on Qiskit Aer’s `AerSimulator` with a default of 4096 shots, yielding an empirical probability distribution over all measured bitstrings.

Acceptance and ranking. A shift offset is considered a match if the ligand bitstring’s measured probability exceeds the uniform baseline $1/N$ — indicating that the oracle and diffusion step measurably amplified that state above what a uniform superposition would produce. Matching windows across all shift offsets are collected into a candidate list carrying: shift offset, start index in `flat_chain`, interactivity score (aggregate h/hb intensity), residues spanned, matched bitstring, measured probability, threshold $1/N$, and number of unique states at that offset. Candidates are sorted by descending interactivity score, so the most chemically interactive matching window is reported first.

Scalability. Oracle and diffusion operators are synthesised from dense $2^{n_{\text{qubits}}} \times 2^{n_{\text{qubits}}}$ unitary matrices. Qiskit’s generic unitary-synthesis routine does not scale gracefully past approximately 10 qubits — in practice, circuit compilation takes tens of seconds to minutes per shift offset beyond this limit. For this reason, `-ligand-max-sites` defaults to 3 (6 qubits); increasing it substantially raises circuit-compilation time and is not recommended without access to more efficient decomposition routines.

3.11 Validation Protocol

The pipeline is validated on crystallographic holo structures, PDB entries in which a ligand is already bound to the protein at its known binding site, as determined experimentally by X-ray crystallography. In this setup the experimental coordinates of the ligand provide an unambiguous geometric ground truth: they tell us exactly where in 3D space the true docking site is located.

For each target, the binding pocket is pre-extracted as the set of protein residues within a fixed spatial radius of the co-crystallised ligand using `make_pockets.py`, and the flat pharmacophoric chain is built on this pocket as described in the preceding sections. Critically, the pipeline operates without any knowledge of the ligand position, it sees only the sequence of pharmacophoric sites of the pocket residues. The ligand is processed separately through the same feature extraction stage to produce its own pharmacophoric profile, and Grover search then compares this profile against all sliding windows of the protein chain to identify the windows whose encoded signature best matches the ligand’s.

The validation criterion is geometric: a prediction is considered correct if the centroid of the winning window, computed as the arithmetic mean of the 3D coordinates of its pharmacophoric sites, which are preserved verbatim from the extraction stage, is close to the true ligand centroid computed from the experimental HETATM coordinates. When the asymmetric unit contains multiple bound copies of the same ligand, the reported metric is the minimum Euclidean distance to any valid copy, so that a prediction correctly targeting a secondary binding pocket is not penalised.

Evaluation is carried out on 29 of the 31 pockets in the benchmark set for which a matching full structure is available (1a08 and 1ddm lack it), spanning diverse protein families and binding-site geometries: 1GM8, 1GPK, 1HNN, 1HQ2, 1HSG, 1IG3, 1K3U, 1L2S, 1MMV, 1N2V, 1N46, 1OPK, 1OYT, 1Q1G, 1R58, 1R9O, 1S19, 1STP, 1T46, 1TZ8, 1U1C, 1XM6, 1Y6B, 1YGC, 1YV3, 1YWR, 2BSM, 3PTB and 4DFR.

Because raw distances in Å are not comparable across pockets of different sizes, an 8 Å error may be near-random in a large pocket but acceptable in a small one, every candidate is also reported as a percentile rank within the distribution of all sliding windows of the same pocket, where 0% denotes the closest window physically available in the representation and 50% corresponds to chance. This normalisation is necessary but not sufficient as a reference, since the pipeline returns several candidates and reporting the best of them is a minimum-of- n statistic: for n windows drawn uniformly at random the expected percentile of the best is $\mathbb{E}[\min] = 1/(n+1)$, which is the baseline against which the “best of candidates” row in the results tables must be read, while the top-1 row is read against the 50% chance level.

Both variants of the search are evaluated under the same protocol. In the *exact* regime a window matches when its first-encoding bitstring equals the ligand’s, as prescribed by the reference paper. In the *amplitude* regime the equality constraint is replaced by a Euclidean distance in the second-encoding amplitude space. Both regimes are evaluated classically: the set of windows satisfying the oracle is fully determined by the encoding before any circuit is executed (see Section 6), so the classical measurement is exact and not an approximation of the quantum run. The absolute value of that 0% reference, how close to the ligand the representation lets any search get at all, is reported in Section 5.1, where it establishes that the 5 Å criterion is attainable.

4 Implementation

4.1 Repository Structure

```
Q-MODE-project/
|-- qmode/
|   |-- grover/
|   |   |-- evaluate.py           # Distance-to-ligand validation
|   |   |-- search.py            # Oracle, diffusion, circuit execution
|   |   |-- __init__.py
|   |-- abraham_hbond.py         # Abraham hb intensity lookup
|   |-- feature_extraction.py     # RDKit pharmacophore features + Crippen h
|   |-- ligand_reader.py         # Ligand HETATM parsing (CCD template / geometric
|   |   fallback)
|   |-- pdb_reader.py            # PDB parsing -> residues, real 3D coords
|   |-- quantum_encoding.py      # First/second quantum encoding
|   |-- qubit_chain.py           # Sliding-window segmentation + qubit chain
|   |-- site_dedup_centroid.py   # Near-duplicate site merging with local centroid
|   |-- site_selection.py        # Topological (BFS) site ordering
|   |-- surface_filter.py        # SASA-based solvent-exposure filter (Shrake & Rupley)
|   |-- __init__.py
|-- scripts/
|   |-- make_pockets.py          # Downloads sample PDBs, crops binding pockets
|   |-- plot_residue_chain.py    # Single-residue 3D + 1D visualization
|   |-- run_pipeline.py          # Main CLI entry point
|   |-- validate_step1.py        # Step 1 validation: template matching and skipped
|   |   residues
|   |-- validate_step2.py        # Step 2 validation: pharmacophore type coverage
|   |-- validate_step3.py        # Step 3 validation: intensity ranges and ranking
|   |-- validate_step4.py        # Step 4 validation: deduplication correctness
|   |-- validate_step5.py        # Step 5 validation: BFS ordering determinism
|   |-- visualize_chain.py       # Flat chain 3D visualization per residue
|   |-- visualize_features.py    # Before/after SASA filter HTML visualization
|-- tests/
|   |-- test_grover_search.py
|   |-- test_ligand_reader.py
|   |-- test_pipeline.py
|-- data/
|   |-- processed/               # JSON/CSV pipeline outputs
|   |-- raw/                     # Input PDB files (pocket and full protein)
|-- README.md
|-- requirements.txt
'-- setup.py
|   |-- percentile_eval.py       # Tables 5/7/8: ceiling/chance, percentiles, k sweep
|   |-- diagnose_descriptor.py   # Section 5.4: Spearman rho, amplitude space vs 3D
|   |-- fast_eval.py             # Benchmark driver over the 29 pockets
```

```
| | -- make_demo_figure.py          # Ceiling-chance figure from percentile_k3.txt
```

4.2 Dependencies

Package	Minimum version
rdkit	$\geq 2023.3.1$
numpy	≥ 1.24
pandas	≥ 2.0
matplotlib	≥ 3.7
biopython	≥ 1.79
qiskit	≥ 2.2
qiskit-aer	≥ 0.17

4.3 Command-Line Interface

The pipeline is exposed through a single script, `scripts/run_pipeline.py`:

```
python scripts/run_pipeline.py --pdb data/raw/3ptb_pocket.pdb --plot
python scripts/run_pipeline.py --pdb data/raw/3ptb_pocket.pdb \
    --output data/processed/ --save-plot data/processed/pocket.png
python scripts/run_pipeline.py --pdb data/raw/4dfr_pocket.pdb \
    --ligand-pdb data/raw/4DFR.pdb
```

Option	Default	Description
<code>-pdb</code>	required	Path to the pocket (or whole-protein) PDB file.
<code>-output</code>	None	Directory for JSON/CSV output files.
<code>-plot</code>	False	Show an interactive visualization.
<code>-save-plot</code>	None	Save plots as PNG images.
<code>-no-surface-filter</code>	filter on	Disable the SASA solvent-exposure filter.
<code>-sasa-threshold</code>	1.0	SASA threshold ($\text{\AA}^2$) for solvent exposure.
<code>-ligand-size</code>	3	Window size (in sites) for the classical sliding-window quantum-chain segmentation (Step 8). This is independent of the real ligand used by Grover search, whose size is derived from <code>-ligand-pdb</code> .
<code>-ligand-pdb</code>	None	Path to a PDB file containing the ligand HETATM records (e.g. the full structure downloaded from the PDB). If provided, enables Grover search for candidate docking sites.
<code>-ligand-code</code>	None	Three-letter ligand code (e.g. MTX) to disambiguate when <code>-ligand-pdb</code> contains multiple non-water HETATM groups. Default: selects the group with the most heavy atoms.
<code>-ligand-max-sites</code>	3	Maximum number of ligand pharmacophore sites used by Grover search (default 3, i.e. up to 6 qubits). Beyond approximately 5 sites (10+ qubits) oracle and diffusion synthesis becomes very slow; increase only if longer runtimes are acceptable.
<code>-max-rows</code>	None	Truncate the flat-sequence and quantum-encoding console tables to their first N rows.

4.4 Output Artifacts

Running the pipeline with `-output` produces up to three artifacts:

`pocket_chain.json` / `pocket_chain.csv` The full flat pharmacophoric chain with per-site metadata: residue label, pharmacophore type, intensity, and 3D coordinates from the crystal structure. The CSV format is convenient for inspection and downstream analysis; the JSON format includes additional metadata such as the pocket centroid and total site count.

`quantum_chain.json` Sliding-window segments of the flat chain, each carrying its first-encoding bitstring $|q_h^{(1)} q_{hb}^{(1)} \dots q_h^{(k)} q_{hb}^{(k)}\rangle$ and the per-site second-encoding amplitude pairs (a, b) and (c, d) . One segment is produced per window position; at the default $k = 3$ sites, the 1a08 pocket produces 133 segments over 135 sites.

`grover_search.json` (*only when `-ligand-pdb` is given*) Grover search candidates ranked by descending interactivity score, each carrying: shift offset, start index in the flat chain, residues spanned, matched bitstring, measured probability, acceptance threshold $1/N$, and distance to the true ligand centroid.

5 Results

This section reports the measurements only; their interpretation is deferred to Section 6. Results are reported across three dimensions: spatial accuracy of the predicted docking site (Section 5.1), robustness across different configurations, and geometric quality of the flat-chain representation (Section ??). A successful prediction is defined here as a candidate window whose centroid falls within 5 Å of the true ligand centroid, a threshold commonly used in docking benchmarks to indicate a near-native pose.

5.1 The search does not localise the binding site

At the default configuration ($k = 3$ sites, 6 qubits), 23 of the 29 pockets produce candidates. The 6 that do not all fail for the same reason: no window clears the uniform $1/N$ probability threshold. Pockets that do produce candidates produce few: 2.43 on average, never more than 3.

The median distance from the top-ranked candidate to the true ligand centroid is 7.75 Å, and 2 cases fall within a 5.0 Å threshold (Table 4). Four of the 23 are pockets in which no window reaches that threshold at all, so the success rate is properly read as 2/19; this is established at the end of the present subsection. Table 3 normalises these distances against the pocket-specific random baseline.

Table 3: Percentile ranking of the Grover candidate windows relative to all possible windows of the same pocket ($k = 3$). Lower is better; 50% is chance.

Metric	Mean percentile	Median	Random baseline
Top-1 by interactivity score	47.2%	42.0%	50.0%
Best of the candidates returned	32.2%	26.2%	30.8%

Two observations follow directly from the table. First, the candidate *set* is statistically indistinguishable from drawing the same number of windows at random: 32.2% against a 30.8% baseline. Second, the top-1 candidate is *worse* than the best of the candidates returned (47.2% vs. 32.2%), which means that ranking by aggregate h/hb interactivity does not merely fail to

help, it actively degrades the selection. The effect is large on individual targets: 1OPK is ranked at the 91.7th percentile while its own best candidate sits at the 8.3rd, 1N46 at the 92.1th against the 14.6th, and 4DFR at the 97.8th against the 33.0rd.

Table 4: Minimum Euclidean distance between the true ligand centroid and the top-ranked candidate window ($k = 3$) across the 29 evaluated protein–ligand pairs. *Failed (Threshold)* indicates that no exact bitstring match cleared the $1/N$ amplitude threshold. Four targets — 1OPK, 1S19, 1Y6B and 1YWR — have no window within 5 Å of the ligand centroid available at all (Table 5), so the success rate over this table is 2/19 rather than 2/23; a dagger marks a target for which no window within 5 Å exists, so the criterion could not have been met by any search, and the distance reported for it measures the search but not its success. 10YT, at 5.0005 Å, falls just outside the threshold and is counted as a failure.

Target	Top-1 distance (Å)	Target	Top-1 distance (Å)
1GM8	9.74	1R90	7.76
1GPK	7.40	1S19 [†]	8.64
1HNN	<i>Failed (Threshold)</i>	1STP	<i>Failed (Threshold)</i>
1HQ2	6.93	1T46	9.00
1HSG	6.48	1TZ8	<i>Failed (Threshold)</i>
1IG3	4.26	1U1C	6.52
1K3U	7.62	1XM6	<i>Failed (Threshold)</i>
1L2S	6.49	1Y6B [†]	7.75
1MMV	9.21	1YGC	22.18
1N2V	<i>Failed (Threshold)</i>	1YV3	5.10
1N46	12.08	1YWR [†]	8.22
1OPK [†]	11.93	2BSM	11.20
10YT	5.00	3PTB	6.81
1Q1G	4.50	4DFR	14.23
1R58	<i>Failed (Threshold)</i>		

Is the criterion reachable? A distance of 7.75 Å is only meaningful once it is known how close a correct answer could have come. The ligand centroid lies in the middle of the cavity, in empty space, whereas pharmacophoric sites lie on the atoms that line it, so there is a floor below which no window can fall. If that floor were above 5 Å the criterion of Table 4 would be unreachable by construction and the benchmark would be measuring pocket geometry rather than the descriptor. We therefore computed, for every pocket, the distance to the ligand centroid of the *closest window that exists at all*, the ceiling on any possible search, and of the *median window*, which is the chance level expressed in Å rather than as a percentile. Both quantities range over all unit-stride windows, the same population against which Table 3 normalises, and neither involves the oracle, the acceptance threshold or the ranking; they are properties of the representation alone. Table 5 reports them.

The criterion is therefore fair: in 25 of the 29 pockets a window within 5 Å of the ligand centroid is physically present in the representation, and the median target sits at 3.76 Å. The four exceptions, 1OPK, 1S19, 1Y6B and 1YWR, whose ceilings lie between 5.01 and 5.67 Å, are large pockets in which no window of three sites reaches the centre of the cavity, and they are the only cases where a failure at 5 Å carries no information about the descriptor. All four returned a candidate and are therefore counted among the 23 pockets of Table 4; excluding them leaves 19 pockets on which the criterion could in principle have been met, and the success rate is properly read as 2/19 rather than 2/23.

The table also restates the result of Table 3 in absolute units, which is how it is most easily read: the target was at 3.76 Å, drawing a window at random gives 7.88 Å, and the search returns

Table 5: Reachability of the 5 Å criterion at $k = 3$, over all unit-stride windows of the 29 evaluated pockets. *Best window* is the closest window that exists in the representation; *median window* is the chance level in Å. Medians are taken across pockets.

Window	Median	Range across pockets
Best available in the representation (ceiling)	3.76 Å	1.21–5.67
Median window of the pocket (chance)	7.88 Å	6.39–10.99
Top-1 returned by the search (Table 4)	7.75 Å	4.26–22.18
Pockets where a window within 5 Å exists		25/29
of which the search returns a candidate		19
of which the candidate falls within 5 Å		2

7.75 Å. Of the 4.12 Å separating chance from the best answer available, the search recovers 0.13 Å — a difference of no consequence against a benchmark whose per-pocket chance level alone ranges from 6.39 to 10.99 Å. The search performs as though it were drawing windows uniformly, and it does so on pockets where a far better answer demonstrably exists.

5.2 Neither the window size nor the matching regime recovers the signal

Because the matching set is determined classically, its quality can be measured directly, independently of any ranking applied afterwards. Table 6 reports the sweep over the window size k in the exact regime.

Table 6: Effect of the sliding-window size on exact-bitstring-match coverage and on the spatial percentile of the matching set.

Max sites (k)	Coverage	Median match percentile	Best of set	Baseline
3	23/29	48.2%	16.8%	18.4%
4	12/29	49.7%	11.7%	23.9%
5	7/29	31.1%	9.8%	39.9%

At $k = 3$ the matching set sits at the 48.2th percentile, exactly chance. Since a ranking function reorders a set without adding information to it, this bounds what any ranking criterion whatsoever could achieve at the default configuration. Raising k to 5 does produce a measurable signal (9.8% against a 39.9% expected baseline), but coverage collapses from 23 to 7 pockets. The cause is informational rather than chemical: the bitstring takes 4^k values, so at $k = 3$ its 64 codes are spread over the 100–600 windows of a typical pocket and collisions are frequent, whereas at $k = 5$ the 1024 codes make an exact match rare, rarer, in most pockets, than the ligand’s own code occurring at all.

Replacing exact equality with a distance in the second-encoding amplitude space decouples the two effects. At $k = 5$ this restores coverage from 7/29 to 29/29 pockets, a genuine and worthwhile gain, but leaves accuracy unchanged: the top-1 candidate lands at the 44.8th percentile (median 42.1th), against 50% for chance, and the best of three candidates at the 27.8th against 25% expected from three random draws. Coverage and accuracy are therefore governed by different bottlenecks, and only the first is a property of the matching criterion.

5.3 The descriptor carries no information about ligand position

The decisive measurement removes the search entirely and asks whether the descriptor and the ground truth are related at all. For every window of every pocket we computed its distance from

the ligand in second-encoding amplitude space and its true 3D distance from the ligand centroid, and correlated the two by Spearman’s rank coefficient (for $K = 3$):

Mean ρ across pockets	−0.008
Median ρ	+0.013
Pockets with $ \rho > 0.3$	2/29
Pockets with the correct sign	16/29

The correlation is absent rather than weak, and the sign is correct in half the pockets, as expected from noise. Two windows that encode similarly are no more likely to be spatially close to each other, or to the ligand, than two windows chosen at random. Since this quantity is measured over all windows and involves neither the oracle, nor the acceptance threshold, nor the ranking, it is a property of the representation alone.

6 Why the Search Does Not Localise the Binding Site

Section 5 establishes that the pipeline does not correctly identify the sub-region of the binding pocket where the ligand binds. A negative benchmark result is only useful to the extent that its cause can be attributed precisely. Three components could in principle be responsible: the classical stages that build the flat-chain representation, the encoding that reduces a window of that representation to a quantum state, or the search that queries the encoding. The following argues that responsibility lies with the encoding, specifically, with what the descriptor can and cannot express about a protein surface. We first eliminate the explanations the benchmark rules out, then characterise what the descriptor does and does not represent, and finally draw out what the result does and does not say about the quantum formulation. The conclusion is constructive: it identifies what a representation would need to express in order for the Grover formulation to have something meaningful to search.

6.1 Eliminating the alternatives

Five explanations present themselves naturally for the failure of Section 5. The following rules out four of them in turn, leaving the encoding as the only remaining candidate.

Bad ranking. The most immediate explanation is that the candidates are ranked badly, and the benchmark appears at first to support it, since the top-ranked candidate is on average worse than the mean of the candidates the pipeline itself returns (47.2th percentile against 32.2rd, Table 3). The sweep of Table 6 shows that this reading is mistaken. At $k = 3$ the set of windows matching the ligand’s bitstring sits at the 48.2th percentile — exactly chance — and a ranking function reorders a set without introducing information the set does not contain. The expected accuracy of any criterion applied to that set is therefore that of a random draw, and the poor showing of `interactivity_score` should not be read as a defect of that particular score: at chance level, a deterministic ordering is as likely to be anti-correlated with the truth as correlated with it. Ranking is not the bottleneck because there is nothing to rank.

Bad matching criterion. Neither is the matching criterion responsible, although it is responsible for a different failure that the same experiments separate out. Replacing exact bitstring equality with a Euclidean distance in second-encoding amplitude space raises coverage from 7 to 29 of 29 pockets at $k = 5$, which is a real gain and the one countermeasure of the two that worked, yet it leaves accuracy at the 44.8th percentile against 50% for chance (Section 5.2). The window size behaves in the same way: the improvement visible at $k = 5$ in the exact regime (9.8% against a 39.9% baseline) is bought with the collapse of coverage that motivated the amplitude regime in the first place, and does not survive once coverage is restored. Coverage and

accuracy are thus governed by different bottlenecks, and only the former belongs to the matching criterion.

Implementation defects or pathological targets. Nor is the result an artefact of a few pathological targets or of the implementation defects we found by validating our own pipeline. The Spearman correlation between amplitude-space similarity to the ligand and true spatial distance exceeds $|\rho| > 0.3$ in only 2 of 29 pockets, and its sign is correct in 16 of 29, half, as expected from noise (Section 5.3), so the outcome is uniform rather than an average over successes and failures. The implementation defects are more substantial in principle and are worth fixing on their own merits. They perturb the values carried in the hb channel, however, and perturbing the values on a channel degrades a correlation rather than moving one from an informative value to $\rho \approx -0.008$.

An unattainable criterion. Nor does the benchmark ask for something the representation cannot deliver. The ligand centroid lies in the empty volume of the cavity while pharmacophoric sites lie on the atoms that line it, so there is a floor below which no window centroid can fall, and if that floor were itself above 5 Å then Table 4 would be recording pocket size and nothing else. Table 5 rules this out. The closest window that exists sits at a median 3.76 Å from the true centroid and is within 5 Å in 25 of the 29 pockets, so on all but four targets a correct answer was present in the candidate set and the search did not select it. The same measurement fixes the scale of the failure in absolute units: the attainable best is 3.76 Å, chance is 7.88 Å, and the search returns 7.75 Å.

3D $\rightarrow$ 1D flattening. The remaining explanation, and the most intuitive one, is that the 3D $\rightarrow$ 1D flattening destroys the spatial arrangement of the sites, so that a window of consecutive positions in the chain is a chemically arbitrary collection with no counterpart on the protein surface. This account has the advantage of pointing at the pipeline’s central modelling decision, but the measurements do not sustain it. At $k = 5$ the median window span is 7.57 Å with a median radius of gyration of 3.06 Å, and only 26.1% of windows exceed 10 Å: consecutive sites form compact local patches of a size comparable to a small ligand, which is to say that chain assembly and topological ordering largely succeed at what they were designed to do. Much of the geometric information is consequently still present, implicitly, in the windows the search examines, and the descriptor fails on objects that are spatially well formed. An explanation of the failure must therefore be sought in what a window *is described by*, not in how it is *assembled*.

6.2 Composition is not a binding signature

What survives the eliminations above is that a window descriptor records the local pharmacophoric *composition* of a patch of protein surface — that is, which interaction families are present in that region, and how strongly. This is analogous to describing a text by the frequency of its letters rather than by the words it contains: the description is accurate but not discriminative. The difficulty is that protein surfaces are compositionally close to uniform, and the report’s own extraction statistics confirm this.

HBondDonor and HBondAcceptor occur in 94–100% of residues, being contributed by the backbone amide nitrogen and carbonyl oxygen that every amino acid carries; Hydrophobe occurs in 79–84%. Those figures were reported in Section 3.2 as evidence that the extraction stage is stable and independent of protein size, which they are, but they carry a second consequence: the same pharmacophoric signature recurs across the entire surface, so a descriptor built from composition assigns similar values to patches that have nothing to do with one another. The search is not looking in the wrong way so much as looking for something that is not rare.

The first encoding compounds this by construction rather than by accident. Reducing each site to two bits gives a window one of 4^k possible values — 64 at the default $k = 3$ — while a single

pocket contributes between roughly 100 and 600 windows, so that many windows necessarily collapse onto the same code and nothing in the construction makes the colliding windows spatially related to each other. The same arithmetic explains the coverage figures of Table 6: increasing k enlarges the code space faster than it enlarges the number of windows, so exact matches become rare without becoming selective, which is why coverage falls to 7 of 29 pockets at $k = 5$ while accuracy improves only marginally.

The same arithmetic applies with more force to the ligand, and there it can be measured directly rather than argued. Encoding the 29 benchmark ligands at $k = 3$ produces only 9 distinct bitstrings of the 64 the code space allows, and a single code, 010101, covers 11 of the 29 — 38% of a benchmark set assembled specifically for chemical diversity. The distribution carries 2.74 bits of the 6 available. For those eleven targets the oracle poses an identical query, namely three consecutive sites polar above threshold and apolar below, which a protein surface satisfies over a large fraction of its windows. This is consistent with the intensity coverage gap recorded in Section 7: the Abraham lookup is indexed by residue and atom name, so it covers protein residues and not ligand atoms, and the heuristic fallback recognises only four functional groups, leaving most ligand atoms at the neutral default and binarising them identically. The premise of the oracle fails as a direct consequence: the ligand’s code equals the code of the window that actually contains it in 1 pocket of 29, against the 0.45 expected if the two were unrelated.

A further loss is structural rather than statistical, and it bears directly on the oracle’s premise. The reduction to two channels described in Section 3.8 maps `HBondDonor`, `HBondAcceptor`, `PosIonizable` and `NegIonizable` onto the single *hb* channel, so that a donor and an acceptor and equally a cation and an anion, become indistinguishable in the descriptor.

6.3 Implications for the quantum stage and for the representation

Before drawing practical consequences, it is important to clarify what this result does and does not say about the quantum formulation itself.

It matters, for how this result should be cited, that none of the preceding is an argument about Grover search. The oracle $\hat{O} = I - 2|x_{\text{lig}}\rangle\langle x_{\text{lig}}|$ implements equality against a fixed bitstring, so the set of windows satisfying it is determined by the encoding before any circuit is prepared, and amplitude amplification alters which element of that set is measured but never the membership of the set itself. The quantum stage accelerates a lookup; it does not score candidates, and it can neither add nor remove spatial accuracy. The negative outcome reported here is therefore not evidence that a quantum formulation of docking-site identification is unproductive. The quadratic speed-up is real, and it is applied to a dictionary whose keys do not distinguish binding sites from the rest of the surface. Properly speaking, the search stage remains untested, since an algorithm cannot be assessed on a problem in which every candidate is equally good.

The eliminations of Section 6.1 also constrain what a redesign would have to supply, and it is here that the negative result acquires practical content. What distinguishes a binding pocket from an arbitrary patch of surface is not the arrangement of a handful of sites but the *enclosure* around them: a pocket is a concavity, and ligandability is a property of the empty volume a ligand could occupy, which no list of sites and intensities expresses at any window size. The report in fact names this quantity already, in another context. Section 3.4 observes that the SASA filter measures solvent accessibility globally, without distinguishing an atom facing the interior of a pocket from one exposed to bulk solvent on the outer surface. Recorded there as a limitation of one filter, the observation is better read in the light of Section 5 as naming what the representation as a whole omits: the pipeline describes where matter is, and never where a ligand would fit.

An interaction-space encoding able to support the search of the reference paper would therefore need, at minimum: a per-site or per-region measure channels that keep the hydrogen-bonding and ionic families distinct, so that complementarity between ligand and protein is expressible at all; and candidate regions assembled from spatial neighbourhoods rather than from sequence

windows, so that residues forming opposite walls of the same pocket can enter a single descriptor. Each of these is compatible with the two-step encoding and with the same Grover formulation downstream. This is the sense in which the present result is a *specification* rather than a dead end: it does not invalidate the scheme proposed by Lillipoulos et al., but identifies precisely what must be placed in the interaction space before that scheme has anything meaningful to search.

7 Known Limitations

This section collects the problems we know about in the current implementation. We group them into two parts: problems that come from reading real PDB files, and limitations of the model and of the libraries we used. None of them explains the benchmark result of Section 6, which Section 6.1 establishes separately.

Template matching ambiguities and missing atoms

Step 1 assigns bond orders by matching each residue against a template. This produces two kinds of diagnostic message across the four test structures.

The first is a warning from RDKit, **More than one matching pattern found**. It appears when a residue contains two atoms that are interchangeable as far as the template is concerned, so there is more than one valid way to match it and RDKit picks one of them arbitrarily. It happens on glutamate, aspartate and arginine. The choice does not matter for us: the two matches differ only in which of the two interchangeable atoms is used, and the pharmacophoric sites we extract afterwards come out identical either way. The number of warnings grows with the size of the protein, from 108 on 1a1c (209 residues) to 579 on 1a0t (1239 residues).

The second kind of message concerns residues that matched no template at all and were skipped. Table 7 reports how many.

Table 7: Step 1 validation results across four protein structures. Skipped residues are those where bond order assignment failed because the file does not contain all the atoms the template expects.

Protein	Total residues	Processed	Skipped	Skip rate
1a08	105	99	6	5.7%
1a0t	1239	1239	0	0.0%
1a1c	209	197	12	5.7%
1a46	280	272	8	2.9%

The skip rate goes from 0% to 5.7%. Looking at which residues get skipped, they are the ones whose record in the file is missing some of the atoms the template requires, which is common in experimentally determined structures published without further processing. 1a0t is the largest of the four structures and skips nothing at all, so the failures depend on how complete the input file is and not on the size of the protein or on our code.

Limitations of the model and of the libraries

A related limitation concerns ligand atoms. The intensity lookup table is indexed by residue name and PDB atom name, so it covers protein residues by construction and not ligand atoms. The heuristic fallback recognises only four functional groups — carboxyl, amine, phenol and thiol — and any atom that falls outside these categories retains the neutral default intensity.

On the quantum computing side, Qiskit constructs the oracle and diffusion operators from dense $2^n \times 2^n$ unitary matrices, and its general-purpose unitary synthesis does not scale gracefully past approximately 10 qubits. This limits `-ligand-max-sites` to small values in practice — the default of 3 sites corresponds to 6 qubits and completes in seconds — and it is why the window-size sweep of Section 5.2 is evaluated classically rather than by executing the circuit for every configuration. The SWAP test proposed by the reference paper for the second-encoding evaluation was also not implemented as a quantum circuit; its classical equivalent, a Euclidean distance between amplitude vectors, was evaluated instead and is reported in Section 5.2. Since the quantum stage does not alter which windows are selected (Section 6.3), a circuit implementation would be expected to yield the same accuracy.

8 Conclusion

Q-MODE implements a complete path from a PDB binding pocket to a Grover search over candidate docking windows executed on a quantum circuit, with site intensities taken from measured physical quantities, Crippen contributions for the hydrophobic channel and Abraham scales for the hydrogen bonding channel, rather than from placeholder constants. We validated the classical stages with a set of scripts that check the output of each step: no residue is left without an expected interaction type and none produces zero features, no site is given a negative intensity, and the BFS topological ordering is deterministic (Sections 3.6 and 3.7).. The encoding and search stages reproduce the first encoding, second encoding, oracle and diffusion operator of the reference paper.

The end to end evaluation is negative, and that is the main result of this report. Across 29 benchmark pockets the candidate windows the search returns are no closer to the crystallographic ligand than windows picked at random from the same pocket: at the 32.2rd percentile against a 30.8% random baseline, and the top ranked candidate at at the 47.2th against 50% for chance. The same result in absolute units, which is how it is most easily read: the closest window that exists in the representation is a median 3.76 Å from the true ligand centroid, drawing a window at random gives 7.88 Å, and the search returns 7.75 Å. Of the 4.12 Å that separated chance from the best answer available to it, it recovers 0.13 Å, which is to say essentially none.. That answer was available: in 25 of the 29 pockets a window within 5 Å of the ligand centroid is physically present in the chain, so the failure is not an artefact of pocket geometry or of an unreachable success criterion. Neither of the two countermeasures we tried changes this. A wider window improves the percentile of the matching set but drops coverage to 7 of 29 pockets, and replacing exact bitstring equality with a distance in amplitude space brings coverage back to 29 of 29, which is a genuine improvement and the part worth keeping, while leaving accuracy at chance.

The cause is in the representation, upstream of the quantum stage. A window descriptor records the pharmacophoric composition of a patch of surface, and that composition is close to uniform: the backbone alone puts a donor and an acceptor on 94% to 100% of residues, so the same signature recurs everywhere. The reduction to two channels also merges donors with acceptors and positive with negative groups, which removes the complementarity that decides whether a ligand binds. One thing our measurements do not support is the explanation we expected to find. The failure cannot be blamed on losing the spatial arrangement of the sites when we flatten the pocket into a chain, because the windows turn out to be compact patches, with a median span of 7.57 Å and a radius of gyration of 3.06 Å, so their local geometry is largely preserved and the descriptor fails on windows that are spatially sensible. What the representation never encodes is how enclosed a region is, which is the property that separates a pocket from any other piece of surface, and the one that Section 3.4 already noted a globally computed SASA cannot see.

Because the oracle tests equality and the set of matching windows is fixed classically before any circuit runs, this result constrains the interaction space encoding and leaves the quantum

search itself untested: an algorithm cannot be judged on a problem where every candidate is equally good. Read that way the outcome is a specification rather than a dead end. It says that an encoding meant to support docking site search has to carry some measure of burial or enclosure, has to keep the hydrogen bonding and ionic families separate so that complementarity can be expressed, and has to build candidate regions from spatial neighbourhoods rather than from windows of the sequence. All of these fit inside the two step encoding scheme and the Grover formulation of the reference paper. The implementation defects we found by validating our own pipeline have since been fixed, with the exception of the dropped insertion codes, and the benchmark outcome did not change which is the strongest form the argument of Section 6.1 can take.

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